

Roger KH Lo

Mechatronics Engineer · Electromechanical Systems Integrator

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PROFILE

Final-year Mechatronics Engineering student (Swinburne University of Technology Sarawak) with hands-on experience in industrial automation, robotics, computer vision and embedded systems. Industry exposure to PLC, HMI and SCADA programming on Beckhoff TwinCAT, with on-site integration of Modbus field devices and IoT systems. Demonstrated project leadership as President of the Engineers Without Borders student chapter.

CORE SKILLS

Automation & control	PLC (Omron, Beckhoff TwinCAT), HMI, SCADA, Modbus, EtherCAT, ECAD, switchboard and field-device integration
Robotics & vision	ROS / ROS2, SLAM, YOLO, SIFT, LoFTR, path planning, ABB RobotStudio, camera calibration (FOV, homography)
Embedded & digital	ESP32, FreeRTOS, MQTT, Arduino, VHDL, RTL design, FPGA prototyping (Altera DE1)
Design & manufacturing	SOLIDWORKS (CAD, Simulation, Additive Manufacturing certified), 3D printing, Design for Assembly, gripper and feeder design
Software & networking	Python, MATLAB, Simulink, FreeBSD, OpenSSL, TLS / mTLS, PKI

EXPERIENCE

Automation Engineer Intern

Dec 2025 – Feb 2026

Sunvision Engineering Sdn Bhd · Sarawak, Malaysia (on-site)

- Programmed PLC, HMI and SCADA systems on hardware and software-based Beckhoff interfaces (TwinCAT, EtherCAT fieldbus).
- Carried out ECAD plotting, on-site inspection, cable checking and troubleshooting at industrial sites, including an oil palm mill and its biogas plant.
- Integrated field devices (Modbus etc.) and IoT systems.

Research Assistant: Robot Interactive Workshop Development

Jun – Jul 2025

Swinburne University of Technology Sarawak Campus · Internship

- Developed an interactive robot workshop in Python using speech recognition, computer vision and AI components.

President (Dec 2024 – Mar 2026) and Project Lead (Feb – Dec 2024)

2 yrs 2 mos

Engineers Without Borders, Swinburne Sarawak Student Chapter

- Led the chapter and its community-focused engineering initiatives, including SDG Carnival '25, community surveys (Kampung Sting) and speaker sessions.
- Managed teams and events, representing the chapter alongside faculty and other student club presidents.

Student Mentor, Chumbaka (YIC)

Apr 2024 – Dec 2025

Part-time

- Mentored youth in Arduino and electronics in preparation for Young Innovators Challenge.

Freelance Photographer & Videographer

Feb 2023 – Mar 2026

Self-employed · Hybrid

- Event, sports and studio photography and event videography for clients.

Student Mentor, Physics

2022

Sunway College

SELECTED PROJECTS

Robot-Assisted Large Surface Defect Detection

Mar 2026 – Present

Research, Swinburne University of Technology

- Building a robot-assisted visual scanning system for automated quality control, combining YOLO detection, SLAM, ROS and path planning to inspect large surfaces.

Automated Assembly Cell: Mechanical Pen

Mar – May 2026

- Designed a robotic work cell to assemble a six-part pen, using Design for Assembly, a custom four-zone pneumatic gripper, vibratory bowl feeders and a three-zone safety enclosure.
- Reduced cycle time from 19.83 s to 10.29 s per unit, with an estimated 87.42% ROI and 1.02-year payback.

Vertical Reciprocating Conveyor

Nov – Dec 2025

- Designed and prototyped a VRC lifting 3 kg over 1500 mm at ≥ 300 mm/s: 2:1 cable-pulley hoist, Maxon RE 65 motor with a custom 19:1 two-stage gearbox, ESCON controller, load cell, IR sensor and limit switches.

Ward Monitoring System (Capstone, Embedded Systems)

Sep – Nov 2025

- ESP32 / FreeRTOS nodes publish sensor data over MQTT to a broker, with patient data in MongoDB and a custom web dashboard (LD2410B radar, DHT11, NeoPixel ring).
- Extended the concept in a FreeBSD-MQTT project secured with mTLS, PKI and CRL.

Automated Storage and Dispense System with CV (Capstone, Mechanism Design)

Sep – Dec 2025

- Designed and built a CAD-modelled, 3D-printed storage and dispense mechanism with a load-cell platform and camera.

Vision-guided Robot Practice, FPGA and PLC training

2025 – 2026

- Vision-guided block stacking and peg-in-hole on ABB IRB robots; VHDL designs (ALU, LFSR) on an Altera FPGA; ladder-logic and HMI programming on Omron PLCs.

EDUCATION

Bachelor of Engineering (Honours): Robotics and Mechatronics Engineering

Final Semester

Swinburne University of Technology (Sarawak Campus)

Expected Completion: December 2026

CERTIFICATIONS & LICENCES

- SOLIDWORKS CAD Design Associate (Jun 2024), Simulation Associate (May 2024) and Additive Manufacturing Associate (Apr 2024), Dassault Systèmes.
- Networking Basics, Cisco (Jun 2025).
- Goethe-Zertifikat A1: Start Deutsch 1, Goethe-Institut.
- Malaysian Class D driving licence (JPJ).

A full project portfolio with photos and diagrams is available on request.